Year 9 biology test 1 revision.

Curriculum content	Questions for revision
• Recognise that the nervous system enables humans to react to their surroundings and coordinate their behavior	Identify the different sections of the nervous system? **Central Nervous System and Peripheral Nervous System** Give an example of negative feedback? **Thermoreceptors detect a decrease in body temperature** **Muscles shiver, thereby increasing body temperature** Give an example of positive feedback? **Uterine contractions during labour** **Secretion of breast milk**
• Recognise the 5 senses	What are the five senses? **Sight, Hearing, Taste, Smell, Touch** Identify the five sense organs? **Eyes, Ears, Tongue, Nose, Skin**
• Understand that cells called receptors detect stimuli (changes in the environment)	What are stimuli? **Changes in the environment that the body can detect** Give examples of stimuli? **Light, sound, chemicals, motion, temperature, pain**
• Identify receptors and the stimuli they detect	Match the receptor cell to the sense organ? Sense organ — Receptor Eye — olfactory nerves Ear — different receptors for each stimuli Skin — tastebuds Tongue — rods and cones Nose — hairs in the cochlea What categories of receptors are there?
• Recognise information from receptors passes along cells (neurones) in nerves to the brain.	There are three types of neurones involved in the nervous system. What are they? **Sensory neuron, relay neuron, motor neuron**
• Understand that the brain	What does the CNS compose of?

<table>
<tr><td>coordinates the response.</td><td>Brain and spinal cord</td></tr>
<tr><td>• Understand that reflex actions are automatic and rapid.</td><td>Compare the nervous and endocrine systems?
<table>
<tr><th>Feature</th><th>Endocrine system</th><th>Nervous system</th></tr>
<tr><td>Speed of message</td><td>Slow</td><td>Fast</td></tr>
<tr><td>Speed of response</td><td>Usually slow</td><td>Immediate</td></tr>
<tr><td>Duration of response</td><td>Long lasting</td><td>Short</td></tr>
<tr><td>Spread of response</td><td>Usually slow</td><td>Very localised</td></tr>
<tr><td>How message travels through body</td><td>In circulatory system — in bloodstream</td><td>In nervous system — along nerves and across synapses</td></tr>
<tr><td>Types of message</td><td>Hormones (chemicals)</td><td>Electrical impulse and neurotransmitters (chemicals)</td></tr>
</table>
Define reflex?
An action that involves only a few neurons and travels only to and from the spinal cord. There is no conscious thought (brain not involved)</td></tr>
<tr><td>• Understand the role of receptors, sensory neurones, motor neurones, relay neurones, synapses and effectors in simple reflex actions.</td><td>Using the example of someone burning there hand on a hot pan. Explain the process that happens? Draw a reflex arc to show this.

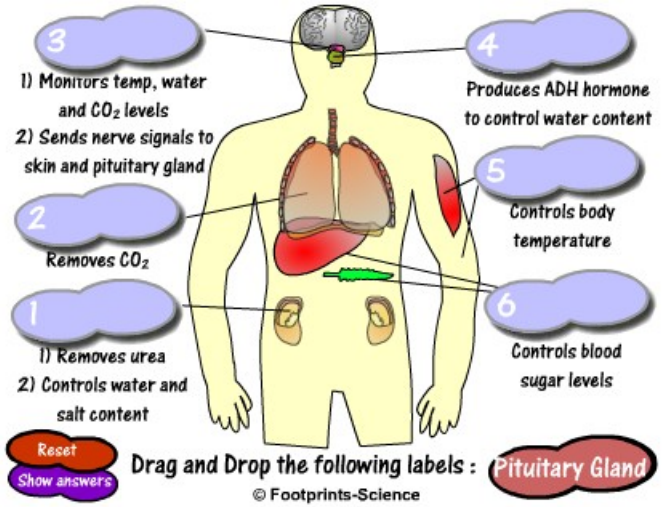
STIMULUS (Heat) → RECEPTOR (Thermoreceptors in hand) → SENSORY NEURON (sends impulse to the CNS) → RELAY NEURON IN SPINAL CORD (interprets the information and sends impulse to motor neurons) → MOTOR NEURON (carries impulse to muscles) → EFFECTOR (Muscles in arm contract) → RESPONSE (Arm moves away from source of heat)

What is a synapse and how does it work? (include the term neurotransmitter)</td></tr>
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	Synapse: A gap between two neurons. When a nervous impulse reaches the end of a neuron, chemicals called neurotransmitters in the first neuron cross the synapse and bind to receptors
• Map and sketch simple reflex actions	Sketch a reflex response model?
• List the internal conditions that need to be controlled in the human body – water content of the body, ion content of the body, temperature, blood sugar levels	The internal conditions of the body are controlled by the process of…… What part of the body is responsible for regulating temperature? What gland is responsible for regulating water content? What gland is responsible for regulating blood sugar levels?
• Recognise that the processes within the body are coordinated by chemical substances	Processes within the endocrine system are controlled by chemical substances called ……? Name some of these chemical substances and what are there function? 1. Thyroxin 2. Adrenalin 3. Aldosterone
• Understand that hormones regulate the functions of many organs and cells	Link the following hormones to their organ; • Testosterone • Progesterone • Growth hormone
• Understand that hormones are secreted by glands and are usually transported to their target organs	How are hormones secreted by glands transported to their target organs?

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• Label the major glands, the hormones they produce and their target organs	
• Identify examples of types of pathogens such as bacteria, viruses, fungi and protozoa and the mode by which pathogens enter the body	What is the definition of a pathogen? How can pathogens be transmitted? Give 2 examples of diseases caused by each category of pathogen? (virus, bacteria, fungi, parasites) Define, what is a vector? What is a host?
• Identify the body's defence mechanisms such as skin, cilia, tears, mucus, gastric juices, lymphocytes and antibodies	First defence line Second defence line Third defence line Explain and give examples of how the different defence lines work?
• Understand the process of immunisation	What is the definition of immunisation? Distinguish between active and passive immunisation?
• Identify a number of infectious diseases and their symptoms	Name three infectious diseases and their symptoms?
• Identify that some diseases are non-infectious and not caused by pathogens such as genetic diseases and lifestyle diseases	Give examples of non-infectious disease and their causes?

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• Identify and discuss how personal choices such as poor diet, lack of exercise, exposure to electromagnetic radiation and high risk factors may contribute to the onset of diseases such as obesity, thrombosis, high blood pressure, heart disease, cancer etc	

Further Questions:

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Looking Back page 174 Questions 1, 3 and 9